

CVT System Model Derivation

Kai Arseneau

February 14, 2026

Contents

1	Overview	3
1.1	Purpose of This Document	3
1.2	Scope	3
1.3	Expected Background Knowledge	3
1.4	Conceptual Overview of a CVT	3
1.5	Modeling Approach	4
2	Notation and Conventions	4
2.1	Coordinate and Sign Conventions	4
2.1.1	Vehicle Longitudinal Direction	4
2.1.2	Rotational Sign Convention	4
2.1.3	CVT Axial Direction	5
2.1.4	Radial Direction	5
2.1.5	Slip Velocity Sign Convention	5
2.2	Symbol Summary	5
3	Units and Dimensions	5
3.1	Base Units	5
3.2	Common Derived Units	6
4	Constants and Parameters	6
5	Modeling Assumptions	6
5.1	Material and Structural Idealizations	7
5.2	Contact and Friction Modeling	7
5.3	Geometric and Kinematic Idealizations	8
5.4	Dynamic and Inertial Modeling	8
5.5	Environmental and Operational Scope	8
6	Governing Principles	9
7	System State Definition	14
7.1	State Vector	15
7.2	Why These Are the State Variables	15
7.3	Numerical Integration Perspective	15

8	Kinematics	16
8.1	Pulley Geometry	16
8.1.1	Primary Effective Radius	16
8.1.2	Secondary Effective Radius via Belt-Length Constraint	16
8.2	CVT Ratio	16
8.3	Ratio Rate of Change	16
9	Pulley Axial Force Models	16
9.1	Primary Axial Force Model	16
9.2	Secondary Axial Force Model	16
9.3	Axial Shift Dynamics	16
10	Belt Forces and Torque Transmission	16
10.1	Axial-to-Radial Force Conversion	16
10.2	Belt Tension Distribution	16
10.3	Maximum Transferable Torque	17
11	Dynamics: No-Slip Baseline	17
11.1	Coupling Torque	17
11.2	Vehicle Longitudinal Dynamics	17
11.3	Final Coupled State-Space System	17
12	Slip Model	17
12.1	Slip Condition and Torque Saturation	17
12.2	Slip Velocity	17
A	Parameter Tables	17

1 Overview

1.1 Purpose of This Document

This document presents a complete dynamic derivation of a continuously variable transmission (CVT) system model. The goal is to develop a physically consistent set of governing equations that describe:

- Rotational dynamics of the primary and secondary assemblies,
- Axial shift dynamics of the primary sheave,
- Belt traction and torque transfer,
- Slip behavior under torque saturation,
- Coupled vehicle longitudinal response.

The resulting formulation provides a structured, state-based model suitable for numerical simulation and performance analysis.

1.2 Scope

This document focuses on the mechanical modeling of the CVT system. It includes force balances, torque transfer mechanisms, and dynamic coupling between rotational and axial motion.

The following are outside the scope of this document:

- Thermal modeling,
- Long-term wear or material degradation,
- Detailed tribological contact mechanics,
- Multi-body vehicle dynamics beyond one-dimensional motion.

All assumptions limiting model fidelity are formally defined in Section 5.

1.3 Expected Background Knowledge

The reader is expected to be familiar with:

- Newton's laws of motion,
- Basic rigid-body rotational dynamics,
- Linear spring models,
- Gear ratios and torque multiplication,
- Basic algebra and differential equations.

Prior familiarity with state-space modeling or numerical integration is not assumed. Relevant concepts are introduced where required.

1.4 Conceptual Overview of a CVT

A continuously variable transmission (CVT) transmits torque between an engine and drivetrain using a belt or chain running between two variable-diameter pulleys.

Each pulley consists of conical sheaves whose axial separation determines the effective belt radius. As the primary sheave shifts axially, the effective radius changes, which in turn modifies the transmission ratio.

Unlike discrete gearboxes, a CVT provides a continuously varying ratio. Torque transfer is governed by belt traction, normal clamping forces, and frictional interaction between the belt and pulley faces.

When transmitted torque exceeds traction capacity, slip occurs. Accurately modeling both stick and slip regimes is essential for predicting performance.

1.5 Modeling Approach

The system is modeled using a minimal set of dynamic state variables, defined in Section 7. All other quantities are derived algebraically from these states.

The derivation proceeds as follows:

- Governing physical laws are defined (Section 6),
- Kinematic relationships between pulley geometry and ratio are derived,
- Axial and radial force models are constructed,
- Belt traction limits are formulated,
- Rotational and axial equations of motion are assembled,
- Slip behavior is incorporated.

The final result is a coupled nonlinear dynamic system suitable for time-domain simulation.

2 Notation and Conventions

2.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

2.1.1 Vehicle Longitudinal Direction

The vehicle is modeled as a one-dimensional system moving along a single longitudinal axis.

- Positive direction ($+x$) is defined as forward vehicle motion.
- Vehicle velocity $v > 0$ corresponds to forward motion.
- Resistive forces (rolling resistance, aerodynamic drag, grade load) act in the negative ($-x$) direction.

Incline Angle If terrain grade is considered, the incline angle α is defined as:

- $\alpha > 0$ for uphill motion.
- $\alpha = 0$ for level ground.

The gravitational component acting along the longitudinal axis is:

$$F_{\text{grade}} = mg \sin \alpha$$

This modifies the longitudinal load only and does not introduce additional spatial dimensions into the model.

2.1.2 Rotational Sign Convention

All angular velocities follow the same convention:

- Positive angular velocity corresponds to forward vehicle motion.
- Primary and secondary pulleys are defined such that $\omega_p > 0$ and $\omega_s > 0$ produce forward belt motion.

- Applied engine torque is positive when it increases ω_p .

Reaction torques opposing motion are therefore negative.

2.1.3 CVT Axial Direction

The axial direction of the primary pulley is defined along its shaft axis.

- Positive axial displacement $s > 0$ corresponds to increasing primary effective radius.
- Increasing s therefore corresponds to an upshift (decreasing overall CVT ratio).

The axial velocity $\dot{s}$ follows the same sign convention.

2.1.4 Radial Direction

Radial direction is defined perpendicular to the pulley axis, measured outward from the center of rotation.

- Radial forces $F_r > 0$ act outward from the pulley center.
- Belt normal forces are treated as positive when compressive.

2.1.5 Slip Velocity Sign Convention

Slip velocity between belt and pulley is defined as:

$$v_{\text{slip}} = r_p(s)\omega_p - r_s(s)\omega_s$$

- $v_{\text{slip}} > 0$ indicates the primary surface speed exceeds the secondary surface speed.
- $v_{\text{slip}} = 0$ corresponds to no-slip contact.

2.2 Symbol Summary

Symbol	Meaning
v	vehicle speed
ω_e	engine angular speed
s	primary axial shift position
$\dot{s}$	primary axial shift velocity
R	CVT ratio (define direction explicitly)

Table 1: Core symbols used throughout the derivation.

3 Units and Dimensions

All quantities in this document are expressed in SI units unless otherwise stated.

3.1 Base Units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 2: SI base units used throughout the model.

3.2 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s ²
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s ²
Force	newton	N
Torque	newton meter	N m
Moment of inertia	kilogram meter squared	kg m ²
Power	watt	W

Table 3: Common derived SI units used in the CVT model.

4 Constants and Parameters

All physical constants, geometric parameters, and calibrated values used in the model are listed below.

Symbol	Name	Value	Unit	Description
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel radius	–	m	Effective rolling radius
G	Final drive ratio	–	–	Secondary-to-wheel speed ratio
I_p	Primary inertia	–	kg m ²	Lumped primary-side inertia
I_s	Secondary inertia	–	kg m ²	Lumped secondary-side inertia
k_p	Primary spring constant	–	N/m	Primary compression spring
k_s	Secondary spring constant	–	N/m	Secondary compression spring
μ	Friction coefficient	–	–	Belt-pulley friction coefficient
β	Wrap angle	–	rad	Belt wrap angle

Table 4: Model constants and parameters.

5 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations

remain valid.

5.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

5.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

5.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Permanent creep and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

5.4 Dynamic and Inertial Modeling

Assumption 9 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 10 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 11 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

5.5 Environmental and Operational Scope

Assumption 12 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 13 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

6 Governing Principles

Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s²)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Bogna Szyk and Steven Wooding (2024).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Encyclopaedia Britannica (2024).

Derivation

Linear spring constitutive relationship in the elastic regime.

Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Dr. Templeman (2024).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}} R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Static Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Wikipedia contributors (2024).

Derivation

Coulomb friction model for impending slip.

Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

Hackaday (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{dT}{T} = \mu d\theta$, then integrate over wrap.

Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m³)
- m is mass (kg)
- V is volume (m³)

Source

The Editors of Encyclopaedia Britannica (2024).

Derivation

Definition of density.

Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m³)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m²)

Source

SoftSchools (2020).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

7 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future value depends on its current value and its time derivative. In numerical

simulation, state variables are the quantities that are directly integrated forward in time at each timestep.

All other quantities in the model (e.g., torque transfer, CVT ratio, belt forces) are computed algebraically from these states.

7.1 State Vector

State Vector

$$\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s).
- ω_s is the angular speed of the secondary pulley (rad/s).
- s is the axial shift position of the primary sheave (m).
- $\dot{s}$ is the axial shift velocity (m/s).

7.2 Why These Are the State Variables

These four variables represent the independent dynamic degrees of freedom of the CVT system:

- The primary pulley rotation (ω_p) evolves according to applied engine torque and belt reaction torque.
- The secondary pulley rotation (ω_s) evolves according to belt torque and external load torque.
- The axial shift position (s) evolves according to the balance of axial forces between the primary and secondary assemblies.
- The axial velocity ($\dot{s}$) is required because axial motion follows a second-order differential equation.

This will be made clear in the following derivations in later sections (TODO: MAKE BETTER). Removing any of these variables would eliminate an independent dynamic degree of freedom from the model.

7.3 Numerical Integration Perspective

At each simulation timestep Δt , the following updates occur:

- ω_p is updated using its angular acceleration.
- ω_s is updated using its angular acceleration.
- s is updated using $\dot{s}$.
- $\dot{s}$ is updated using axial acceleration.

Thus, the state vector evolves according to differential equations derived in later sections.

All other model quantities — including CVT ratio, belt slip velocity, radial forces, and torque capacity — are computed directly from the current values of ω_p , ω_s , and s .

8 Kinematics

8.1 Pulley Geometry

8.1.1 Primary Effective Radius

$$r_p(s) = \dots \quad (8.1)$$

8.1.2 Secondary Effective Radius via Belt-Length Constraint

$$r_s = \dots \quad (8.2)$$

8.2 CVT Ratio

$$R(s) = \frac{r_s(s)}{r_p(s)} \quad (8.3)$$

8.3 Ratio Rate of Change

$$\dot{R} = \frac{dR}{ds} \dot{s} \quad (8.4)$$

9 Pulley Axial Force Models

9.1 Primary Axial Force Model

$$F_{p,ax} = \dots \quad (9.1)$$

9.2 Secondary Axial Force Model

$$F_{s,ax} = \dots \quad (9.2)$$

9.3 Axial Shift Dynamics

Axial Shift Equation of Motion

$$m_{\text{shift}} \ddot{s} = F_{p,ax} - F_{s,ax}$$

10 Belt Forces and Torque Transmission

10.1 Axial-to-Radial Force Conversion

$$F_{r,p} = \dots \quad F_{r,s} = \dots \quad (10.1)$$

10.2 Belt Tension Distribution

Using Section 6,

$$\frac{T_1}{T_2} = e^{\mu\beta} \quad (10.2)$$

10.3 Maximum Transferable Torque

Maximum Transferable Torque

$$T_{\max,p} = \dots \quad T_{\max,s} = \dots$$

11 Dynamics: No-Slip Baseline

11.1 Coupling Torque

Coupling Torque Definition

$$T_c = \dots$$

11.2 Vehicle Longitudinal Dynamics

$$m\dot{v} = \frac{T_{\text{wheel}}}{r_{\text{wheel}}} - F_{\text{resist}} \quad (11.1)$$

11.3 Final Coupled State-Space System

State-Space Form

$$\dot{x} = f(x, u)$$

12 Slip Model

12.1 Slip Condition and Torque Saturation

$$|T_c| \leq T_{\max} \Rightarrow \text{no slip} \quad (12.1)$$

$$T = T_{\max} \text{sgn}(T_c) \Rightarrow \text{slip} \quad (12.2)$$

12.2 Slip Velocity

$$v_{\text{slip}} = r_p \omega_p - r_s \omega_s \quad (12.3)$$

A Parameter Tables

References

- 2024 Power Test, LLC. How to calculate horsepower. Available from the Power Test Dyno website, 2024. URL <https://powertestdyno.com/how-to-calculate-horsepower/>. Accessed: 2024-10-08.
- Bogna Szyk and Steven Wooding. Centrifugal force calculator. Available from the Omni Calculator website, 2024. URL <https://www.omnicalculator.com/physics/centrifugal-force>. Accessed: 2024-10-08.
- Marshall Brain. How gear ratios work. *HowStuffWorks*, 2023. URL <https://science.howstuffworks.com/transport/engines-equipment/gear-ratio.htm>. Accessed: 2024-10-08.
- Dr. Templeman. Spring design formulas. Available from Dr. Templeman’s website, 2024. URL <https://www.drtempleman.com/spring-resources-old/spring-design-formulas-2>. Accessed: 2024-10-08.
- Encyclopaedia Britannica. Hooke’s law. *Britannica.com*, 2024. URL <https://www.britannica.com/science/Hookes-law>.
- Hackaday. Cable mechanism maths: Designing against the capstan equation. Available from the Hackaday website, 2021. URL <https://hackaday.com/2021/01/26/cable-mechanism-maths-designing-against-the-capstan-equation/>. Accessed: 2024-10-08.
- SoftSchools. Air resistance formula. Available from the SoftSchools website, 2020. URL https://softschools.com/formulas/physics/air_resistance_formula/85/. Accessed: 2024-10-08.
- The Editors of Encyclopaedia Britannica. Density. *Encyclopedia Britannica*, 2024. URL <https://www.britannica.com/science/density>. Accessed: 2024-10-08.
- Wikipedia contributors. Friction. Available from the Wikipedia website, 2024. URL <https://en.wikipedia.org/wiki/Friction>. Accessed: 2024-10-08.